

D.A.V. LOHAGHAT
PERIODIC ASSESSMENT - I (2025-26)
SUBJECT - SCIENCE
CLASS-X

MM : 40

Time: 2:00 hours

CHEMISTRY

- Q 1.** The chemical equations are balanced to satisfy one of the following laws in chemical reactions. This law is [1]
- A. Law of conservation of momentum C. Law of conservation of energy
B. Law of conservation of mass D. law of conservation of magnetism
- Q 2.** You are given the solution of lead nitrate. In order to obtain a yellow precipitate you should mix with it a solution of : [1]
- A. potassium chloride C. potassium sulphide
B. potassium nitride D. potassium iodide
- Q 3.** State the various methods for preventing or retarding rancidity of food. [2]
- Q 4.** What do you understand by exothermic and endothermic reactions ? [2]
- Q 5.** Translate the following statements into chemical equations and then balance the equations : [2]
- (a) Hydrogen sulphide gas burns in air to give water and sulphur dioxide.
(b) Phosphorus burns in oxygen to give phosphorus pentoxide.
(c) Carbon disulphide burns in air to give carbon dioxide and sulphur dioxide.
(d) Aluminium metal replaces iron from ferric oxide, Fe_2O_3 , giving aluminium oxide and iron.
- Q 6.** The metal **M** reacts vigorously with water to form a solution **S** and a gas **G**. The solution **S** turns red litmus to blue whereas gas **G**, which is lighter than air, burns with a pop sound. Metal **M** has a low melting point and it is used as a coolant in nuclear reactors. [5]
- (a) What is metal **M** ?
(b) What is solution **S** ? Is it acidic or alkaline ?
(c) What is gas **G** ?
(d) Write a balanced chemical equation for the reaction which takes place when metal **M** reacts with water.
(e) Is this reaction exothermic or endothermic

BIOLOGY

- Q 7.** The kidneys in human beings are a part of the system for- [1]
- A. Nutrition C. Respiration
B. Excretion D. Transpiration
- Q 8.** Which part of the heart receives impure blood from the body? [1]
- A. Right atrium C. Right ventricle
B. Left atrium D. Left ventricle
- Q 9.** Oxygen liberated during photosynthesis comes from [1]
- A. Water C. CO₂
B. Chlorophyll D. Glucose
- Q 10.** Name the stored food of animals. [1]
- Q 11.** What makes the blood appears red in colour? [1]
- Q 12.** Name the largest vein in human body. [1]
- Q 13.** Write the name of excretory organs of the following organisms- [1]
- i. Earthworms ii. Insects
- Q 14.** Write the name of the artery which carries deoxygenated blood [1]
- Q 15.** Explain double circulation in human beings. [2]
- Q 16.** Explain the transport mechanism in plants. [2]
- Q 17.** Describe the structure and function of nephrons. [2]

PHYSICS

- Q 18.** To get an image of magnification -1 on a screen using a lens of focal length 20 cm, the object distance must be: [1]
- A. Less than 20 cm C. 40 cm
B. 30 cm D. 80 cm
- Q 19.** Absolute refractive index of water and glass is $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is [1]
- A. $\frac{9}{4} \times 10^8$ m/s C. $\frac{16}{9} \times 10^8$ m/s
B. $\frac{7}{3} \times 10^8$ m/s D. $\frac{9}{8} \times 10^8$ m/s

Q 20. Mirror '**X**' is used to concentrate sunlight in solar furnace and Mirror '**Y**' is fitted on the side of the vehicle to see the traffic behind the driver. Which one of the following statements are true for the two mirrors? [1]

- The image formed by mirror '**X**' is real, diminished and at its focus.
- The image formed by mirror '**Y**' is virtual, diminished and erect.
- The image formed by mirror '**X**' is virtual, diminished and erect.
- The image formed by mirror '**Y**' is real, diminished and at its focus.

A. (i) and (iv) B. (i) and (ii) C. (ii) and (iii) D. (iii) and (iv)

Q 21. (a) During an experiment, a student observes that the given mirror is forming an erect and diminished image for all positions of the object placed in front of it. Based on this information, answer the following questions: [2]

- Identify the mirror.
- Using new Cartesian sign convention, write the magnitude of the magnification in this case.
- What will be the radius of curvature of the mirror if its focal length is 20 cm?

OR

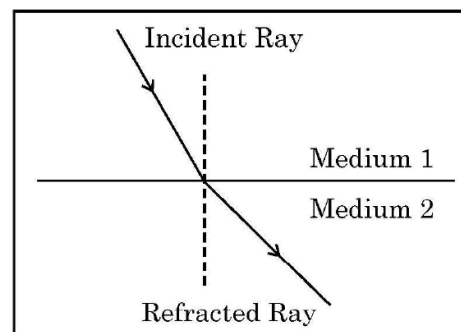
(b) A student wishes to study the image formation by a concave mirror using candle flame as object. State the type of the image formed by the mirror and mention the change in the image formed, if any, that he observes when the candle flame is gradually moved away from the pole of the mirror. Draw a ray diagram to show the image formation when the object distance is nearly equal to the radius of curvature of the mirror. [2]

Q 22. (a) The power of a lens is -1.25 D. Based on this information, find out [2]

- The type of lens and its focal length.
- The numerical values of the focal lengths of two lens **A** and **B** are **10 cm** and **20 cm** respectively. Which one of the two will show higher degree of convergence/divergence? Give reason to justify your answer.

OR

(b) Study the figure in which the path of a ray of light going from Medium 1 to Medium 2 shown. [2]



- Out of the two media- Medium 1 and Medium 2, in which is the speed of light is more?
- State reason of bending of the refracted ray away from the normal.
- Express refractive index of Medium 2 with respect to Medium 1 in terms of speed of light in two media.

- Q 23.** (a) If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer. [3]

OR

- (b) A convex lens forms an 8.0 cm long image of a 2.0 cm long object which is kept at a distance of 6.0 cm from the optical centre of the lens. If the object and the image are on the same side of the lens, find [3]
- the nature of the image
 - the position of the image, and
 - the focal length of the lens.

- Q 24.** (a) Analyse the following observation table showing variation of image distance (v) with object distance (u) in case of a convex lens and answer the questions that follow without doing any calculations: [3]

Observation Number	Object distance (u) in cm	Image distance (v) in cm
1	-15	-60
2	-25	+100
3	-30	+60
4	-40	+40
5	-60	+30
6	-100	+25

- Determine the focal length of the lens. Give reason for your answer.
- Find magnification of the image formed in observation No. **5**
- The numerical value of magnification in case of observation **1** and **2** is same. List **two differences** in the image formed in these two cases.

OR

- (b)
 - State Snell's law of refraction of light. [3]
 - Draw ray diagrams to show the nature, position and relative size of the image formed by a convex mirror when the object is placed (i) at infinity and (ii) between infinity and pole P of the mirror.